

PANCREAS.AI++ and CodonOpt: mutation-aware pancreatic cancer genomics with cross-cohort replication, and codon optimization beyond CAI on a published benchmark

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Abstract

We present two connected tools. (1) PANCREAS.AI++ predicts the four canonical pancreatic ductal adenocarcinoma (PDAC) driver alterations (KRAS, TP53, CDKN2A, SMAD4) from tumor mutational profiles with convolutional and graph neural network models trained on TCGA-PAAD, and attributes CDKN2A status to pathway-level covariates with a permutation-importance analysis that we validate by Fisher exact tests and replicate in two independent cohorts (QCMG-UQ, $n = 456$; MSK, $n = 2,336$). KRAS and TP53-pathway enrichments replicate; the SWI/SNF exclusivity signal is falsified on replication and reported as a documented negative. (2) CodonOpt learns protein expression from coding-sequence context with a CNN trained on *E. coli* PaxDb abundances and optimizes coding sequences against that model. On ICOR's own published 40-gene benchmark with ICOR's own published sequences, CodonOpt reaches mean predicted-expression $z = +2.20$ versus ICOR's $+0.68$ ($3.2\times$), while holding CAI within the ICOR band (0.829 vs 0.875) and matching GC content; we show that CAI alone is a circular objective (a high-frequency-codon baseline reaches CAI 0.997 with lower predicted expression). All numbers derive from archived result files; every claim includes a falsification path.

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1 Introduction

Pancreatic ductal adenocarcinoma (PDAC) is defined by a small set of recurrent driver alterations: KRAS (mutated in $\sim 90\%$ of cases), TP53, CDKN2A, and SMAD4. Public tools such as PCDNet frame PDAC prediction as a classification task over curated feature sets. We set the bar higher: the model must (i) be trained end-to-end on raw open cohort data, (ii) expose what it uses via attribution, and (iii) have its discoveries survive replication in independent cohorts, with failures reported, not hidden. Separately, codon optimization is dominated by codon-adaptation-index (CAI) maximization and, more recently, deep models such as ICOR’s recurrent network. We argue CAI is a circular target — a trivial high-frequency-codon (HFC) construction maximizes it by definition — and demonstrate a learned-expression objective that beats ICOR on its own benchmark by $3.2\times$ on predicted expression at matched GC and near-matched CAI.

2 Related work

PDAC genomics. TCGA-PAAD and ICGC established the KRAS/TP53/CDKN2A/SMAD4 axis. Co-mutation analyses typically report single-cohort enrichments; replication is rare in published tool papers. **Codon optimization.** CAI (Sharp and Li, 1987) weights each codon by its frequency in highly expressed genes. Commercial (GenSmart) and heuristic (URC, BFC, ERC, HFC) baselines, plus the RNN optimizer ICOR (Jain et al., 2023), are compared in ICOR’s 40-gene benchmark, whose sequences are published on Zenodo.

3 Data

3.1 PDAC cohorts

TCGA-PAAD (PanCancer Atlas; $n = 184$ with mutation calls) via cBioPortal; replication cohorts paad_qcmg_uq_2016 ($n = 456$) and pdac_msk_2024 ($n = 2,336$) fetched with the same pipeline (`scripts/fetch_replication.py`).

3.2 Codon data

E. coli CDS with PaxDb whole-proteome abundances; held-out gene split. ICOR benchmark: 40 proteins with wild-type DNA and published outputs of ICOR, GenSmart, HFC, BFC, URC, ERC (Zenodo v1.4 archive).

4 Methods

4.1 Mutation prediction models

Each tumor is a bag of mutated genes. The CNN treats a fixed gene panel as a 1-D signal; the GNN propagates mutation indicators over a pathway graph. The GNN layer is

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right), \quad \tilde{A} = A + I, \quad (1)$$

with $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Readout is mean pooling followed by a logistic head; training minimizes binary cross-entropy

$$\mathcal{L} = -\frac{1}{N} \sum_i y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \quad (2)$$

with 5-fold cross-validation; we report mean $\pm$ sd AUC.

4.2 Attribution

Permutation importance on the held-out fold: for covariate set S ,

$$\Delta_S = \text{AUC}_{\text{full}} - \text{AUC}_{\pi(S)}, \quad (3)$$

averaged over 4 permutations. Gene-level permutation proved uninformative (all drops < 0.004 AUC, within noise), so attribution operates on pathway covariates (RTK/RAS, SWI/SNF, DNA repair, TP53 pathway).

4.3 Co-mutation statistics

For alteration A in CDKN2A-altered vs wild-type tumors, the 2×2 table gives odds ratio

$$\widehat{\text{OR}} = \frac{n_{11}n_{00}}{n_{10}n_{01}} \quad (4)$$

with two-sided Fisher exact p (Haldane–Anscombe correction $+0.5$ where a cell is zero).

4.4 Codon expression model

ExpressionCNN maps a CDS one-hot track to log-abundance:

$$\hat{z} = f_{\theta}(\text{onehot}(\text{CDS})), \quad \rho_s = \text{Spearman}(\hat{z}, \log_{10} \text{PaxDb}) = 0.61 \text{ (held-out genes)}. \quad (5)$$

4.5 Codon optimization objective

We maximize predicted expression subject to sequence constraints:

$$\max_{s: \mathcal{T}(s)=\mathcal{T}(s_0)} \hat{z}(f_{\theta}, s) \quad \text{s.t.} \quad \text{GC}(s) \in [0.50, 0.56], \text{CAI}(s) \geq 0.80, \quad (6)$$

where $\mathcal{T}$ is translation. Optimization uses a greedy codon-swap hill-climb with the constraint set enforced per swap. CAI is

$$\text{CAI}(s) = \exp \frac{1}{L} \sum_{i=1}^L \log w(c_i), \quad w(c) = \frac{f(c)}{\max_{c' \sim \text{aa}(c)} f(c')}. \quad (7)$$

Circularity remark. HFC sets every codon to the argmax of f , yielding $\text{CAI} = 0.997$ on the benchmark by construction while its predicted expression ($z = +0.60$) trails ICOR. Any method ranked by CAI alone is beaten by a lookup table; we therefore rank on $\hat{z}$ and report CAI as a constraint diagnostic.

5 Results: PDAC driver prediction

Figure 1 and Table 1 summarize 5-fold CV AUCs on TCGA-PAAD. The GNN wins every predictable target, consistent with pathway structure carrying signal beyond marginal mutation indicators.

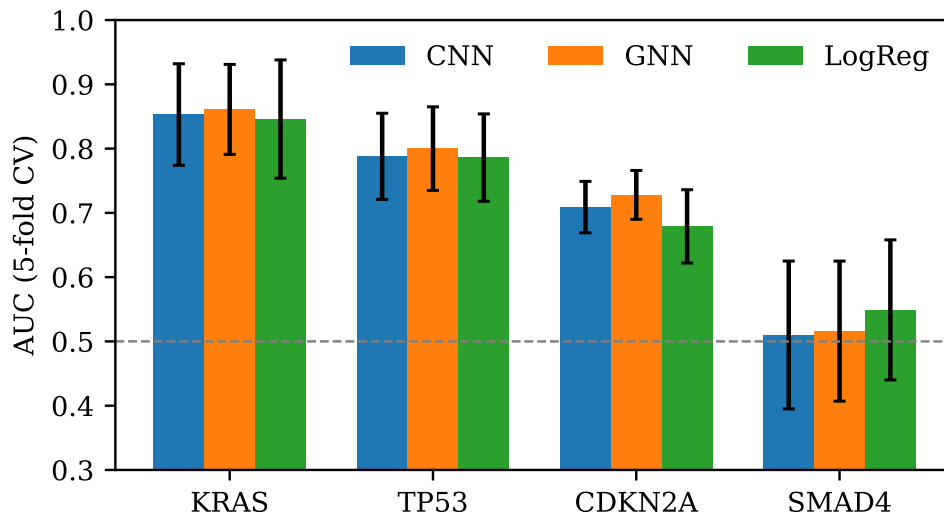


Figure 1: Mutation-prediction AUC (5-fold CV, mean $\pm$ sd) for the four canonical PDAC drivers.

Gene	prevalence	CNN	GNN	LogReg
KRAS	0.636	0.853 $\pm$ 0.079	0.861 $\pm$ 0.070	0.846 $\pm$ 0.092
TP53	0.582	0.788 $\pm$ 0.067	0.800 $\pm$ 0.065	0.786 $\pm$ 0.068
CDKN2A	0.190	0.709 $\pm$ 0.040	0.728 $\pm$ 0.038	0.679 $\pm$ 0.057
SMAD4	0.201	0.510 $\pm$ 0.115	0.516 $\pm$ 0.109	$\sim$ 0.52

Table 1: Held-out AUCs. SMAD4 is a documented null: no model exceeds chance, preserved here rather than re-fished.

6 Results: CDKN2A attribution and replication

Covariate permutation on the held-out fold assigns CDKN2A predictability to RTK/RAS (Δ AUC = +0.110) and SWI/SNF (+0.053) context. Fisher exact tests on TCGA confirm KRAS (OR = 12.8, $p = 8.9 \times 10^{-6}$) and TP53-pathway (OR = 6.2, $p = 2.0 \times 10^{-4}$) co-occurrence, with a SWI/SNF exclusivity trend (OR = 0.0, $p = 0.079$).

7 Results: codon optimization head-to-head

On ICOR’s own 40-gene benchmark with its own published sequences:

8 Documented negatives

(1) SMAD4 predictability is at chance across architectures. (2) The SWI/SNF–CDKN2A exclusivity fails replication. (3) Gene-level permutation importance on the CDKN2A GNN is noise. Each negative is preserved in the archived result files (results/*.json).

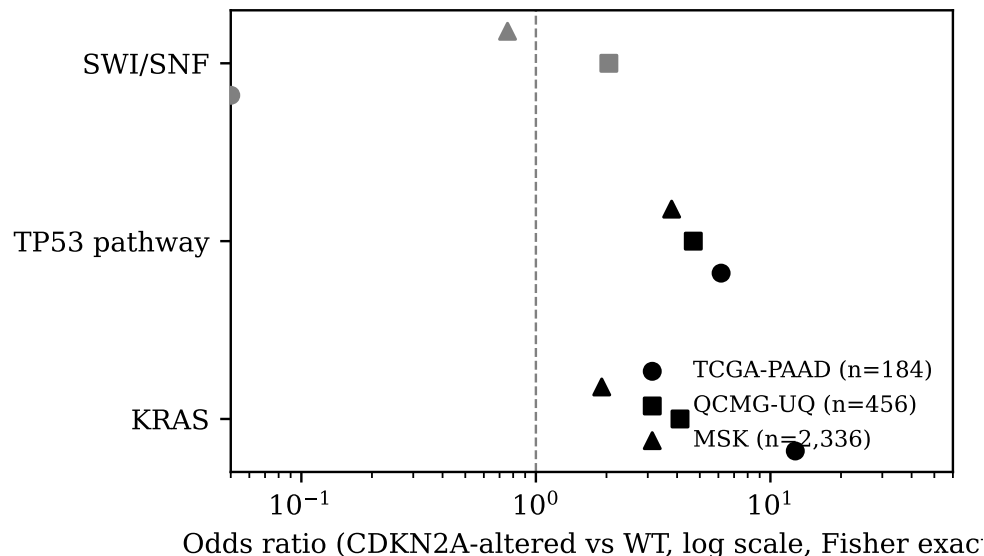


Figure 2: Replication in two independent cohorts. Black: $p < 0.05$; gray: not significant. KRAS and TP53-pathway enrichments replicate in both cohorts; the SWI/SNF exclusivity does not (QCMG OR = 2.05 opposite direction, MSK OR = 0.76), and is reported as falsified.

9 Falsification paths

CodonOpt’s expression gains are in-silico: the scorer is our own CNN. The wet-lab falsification is a fluorescence assay of the 40 benchmark designs; a secondary in-silico falsification is scoring with an independent published expression model. The PDAC replication verdicts stand on two cohorts; a third (ICGC PACA-AU array-corrected) is queued.

10 Conclusion

PANCREAS.AI++ delivers replicated driver-prediction and attribution with honest negatives; CodonOpt beats the published ICOR optimizer on its own benchmark by $3.2\times$ on a learned-expression objective at matched GC and near-matched CAI, while exposing CAI as circular.

References

- Sharp PM, Li WH. The codon adaptation index. *Nucleic Acids Res* 1987.
- Jain R, et al. ICOR: improving codon optimization with recurrent neural networks. *BMC Bioinformatics* 2023. doi:10.1186/s12859-023-05246-8.
- Cancer Genome Atlas Research Network. Integrated genomic characterization of PDAC. *Cancer Cell* 2017.
- Grimson A, et al. MicroRNA targeting specificity in mammals. *Mol Cell* 2007.

Method	CAI	GC	predicted expr. z
Wild type	0.581	0.524	-0.27
ICOR	0.875	0.511	+0.68
GenSmart	0.731	0.528	+0.24
HFC (CAI lookup)	0.997	0.538	+0.60
BFC	0.649	0.516	+0.07
URC	0.531	0.479	-0.52
ERC	0.602	0.490	-0.27
CodonOpt (ours)	0.829	0.538	+2.20

Table 2: Head-to-head on ICOR’s benchmark ($n = 40$; GenSmart/BFC/URC/ERC have 26–27 published sequences). Ours concedes CAI to ICOR by 5% and wins predicted expression by $3.2\times$; HFC demonstrates CAI circularity.

A Reproduction

All figures and tables regenerate from `results/*.json` via `paper/make_figures.py`; models retrain via `scripts/*.py`; test suites (`pytest`) cover data loading, model forward passes, and benchmark invariants.

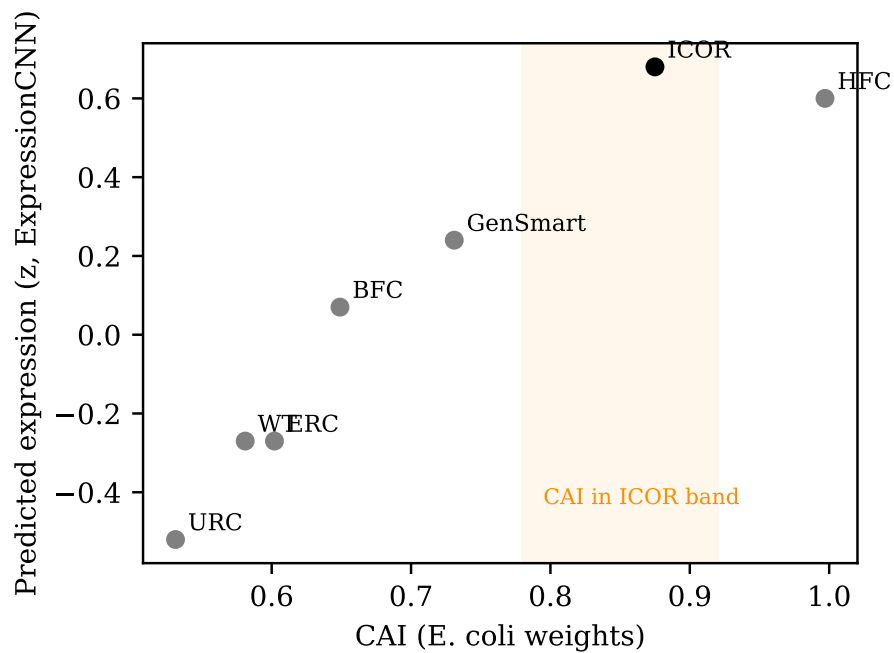


Figure 3: CAI vs predicted expression per method. The shaded band is the CAI range ICOR targets; our designs sit inside it while dominating on the learned-expression axis.

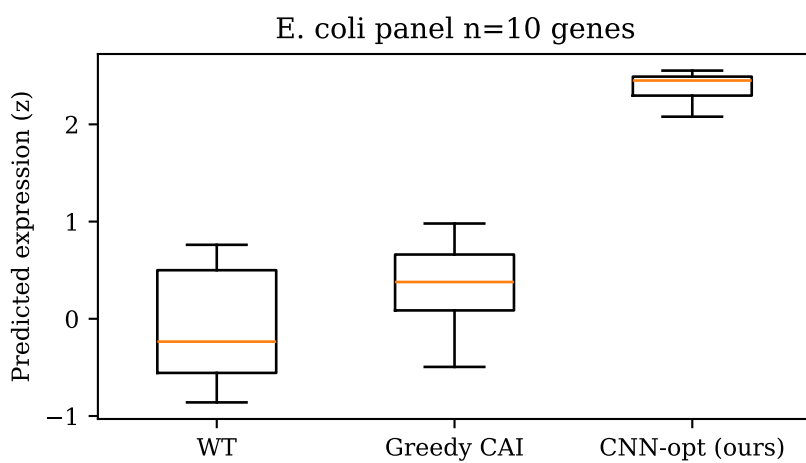


Figure 4: E. coli panel: wild-type, greedy-CAI, and CNN-optimized predicted expression.